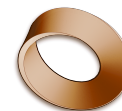


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.4 – Transverse Waves

Unit 1: Oscillations & Waves

Learning intentions

- a) I can explain how transverse waves transfer energy without transferring matter.
- b) I can describe the oscillation direction relative to wave travel in a transverse wave.
- c) I can identify peaks, troughs, amplitude and wavelength on a displacement-distance graph.
- d) I can calculate the wavelength of a transverse wave from a given diagram or description.

Theory

A **wave** is a disturbance that transfers energy from one place to another without transferring matter. In a **transverse wave**, the particles of the medium oscillate (vibrate) at right angles (perpendicular) to the direction in which the wave travels. For example, if a wave moves horizontally along a rope, the particles of the rope move up and down. The particles do not travel with the wave; they simply oscillate about a fixed **rest position**. This means that after the wave passes, the particles return to their original location.

The key features of a transverse wave include the **peak** (or crest), which is the highest point of the wave, and the **trough**, which is the lowest point. The **amplitude** is defined as the maximum displacement of a particle from its rest position. It is measured from the rest position line to the top of a peak or the bottom of a trough. The **wavelength** is the distance between two consecutive corresponding points on the wave, such as from peak to peak or trough to trough. Wavelength is typically denoted by the symbol λ and is measured in metres (m).

To determine these features from a **displacement-distance diagram**, look at the horizontal axis which represents distance. The amplitude is the vertical distance from the central rest line to a peak. The wavelength is the horizontal distance between two adjacent peaks. It is important not to confuse this with a displacement-time graph, where the horizontal axis represents time. In a displacement-distance graph, you can directly measure the physical length of one complete wave cycle.

Consider a wave on a string where the rest position is at 0 cm. If the peak reaches +4 cm and the trough reaches -4 cm, the amplitude is 4 cm. If the first peak is at 2 m and the next peak is at 6 m, the wavelength is calculated by subtracting the positions: $6\text{ m} - 2\text{ m} = 4\text{ m}$. This shows that one complete cycle spans 4 metres.

Complete the sentence: In a transverse wave, the particles oscillate at right angles to the direction of energy transfer, and the _____ is measured from the rest position to a peak.

Word bank: amplitude · wavelength · trough · peak · frequency

Activity

Mapping Transverse Waves

You will need: a long slinky spring or a rope, a ruler, and a partner.

- Tie one end of the slinky or rope to a fixed point (e.g., a door handle or chair leg).
- Hold the other end loosely and keep the medium taut but not tight.
- With your partner, create a single transverse pulse by moving your hand sharply sideways (left or right) once. Observe how the disturbance travels along the length while

the rope itself stays in place.

- Now, move your hand up and down continuously to create a continuous wave pattern.
- Ask your partner to mark the rest position on the floor with tape.
- Measure the maximum height of a peak from the rest position tape (amplitude).
- Measure the horizontal distance between two consecutive peaks (wavelength).

Record your measurements in the table below. Assume the wave is stable and consistent.

Measurement	Value (m)	Notes
Amplitude	_____	Measured from rest to peak
Wavelength	_____	Measured peak to peak
Direction of oscillation	_____	e.g., Up/Down or Left/Right
Direction of wave travel	_____	e.g., Along the rope

Questions

1. Explain in your own words why a floating leaf on a pond bobs up and down as water waves pass but does not drift towards the shore with the wave energy.

2. On a displacement-distance diagram, the rest position is at 0 cm. The highest point of the wave is at +5 cm. What is the amplitude of this wave?

3. A transverse wave has peaks located at 1.2 m, 3.6 m and 6.0 m along the x-axis. Calculate the wavelength of this wave.

4. Describe two differences between the motion of particles in a transverse wave and the direction the wave travels.

5. If you increase the speed at which you shake a rope up and down (keeping the amplitude constant), how does the wavelength change if the frequency increases? (Hint: think about the relationship $v = f\lambda$ qualitatively).

Extension

Research electromagnetic waves. Since they do not require a medium, how can they still be described as transverse waves? Investigate what is oscillating in an electromagnetic wave and why this makes them fundamentally different from mechanical transverse waves like those on a rope or water surface.
